

II- Assay questions.

II-1) In a follow-up to Beadle and Tatum's original study, they decide to identify the genes required for synthesis of another amino acid, Leucine (L). Like Arginine, Leucine is synthesized via a series of steps. Using X-rays you create Leucine auxotrophs. You identify 20 mutants. Complementation tests reveal 4 complementation groups: 1, 2, 3, 4. They suspect that the following molecules may be precursors to Leucine (L): isopropylmalate (IP), ketoisocaproate (KC), dihydroxyvalerate (DV), ketoisovalerate (KV). Each mutant is tested for growth on minimal media (mm) + each Leucine precursor. The results are below:

mutant	mm+ KC	mm+ IP	mm+KV	mm+ DV	mm+Leucine
1	no	grows	no	no	grows
2	no	no	no	no	grows
3	grows	grows	no	grows	grows
4	grows	grows	no	no	grows

Draw out the pathway indicating with arrows the order of each precursor and showing where each gene functions in the pathway. (4pts)

KV—3--- DV---4---KC---1----IP-----2----Leu

II-2) The genetic analysis shown below describe the genotypic ratios of three genes in corn: an ("Anther ear"), br ("brachytic") and f ("fine stripe"). When a plant that is heterozygous for each of these markers is test-crossed with a homozygous recessive plant, the following results are obtained:

Test Cross	Progeny Numbers
wild type	98
fine	26
brachytic	2
brachytic, fine	379
anther	385
anther, fine	3
anther, brachytic	20
anther, brachytic, fine	87
Total Offspring:	1000

- 1) Determine the map distance between the three genes. (3 pts)
- 2) Determine the order of these genes. (1pt)

II-2) The sequence of a segment of mRNA, beginning with the initiation codon, is given here, along with the corresponding sequences from several mutant strains.

Normal AUG ACA CAU CGA GGG GUG GUA AAC CCU AAG...

Mutant 1 AUG ACA CAU CCA GGG GUG GUA AAC CCU AAG...

Mutant 2 AUG ACA CAU CGA GGG UGG UAA ACC CU AAG...

Mutant 3 AUG ACG CAU CGA GGG GUG GUA AAC CCU AAG...

Mutant 4 AUG ACA CAU CGA GGG GUU GGU AAA CCC UAA G...

Mutant 5 AUG ACA CAU UGA GGG GUG GUA AAC CCU AAG...

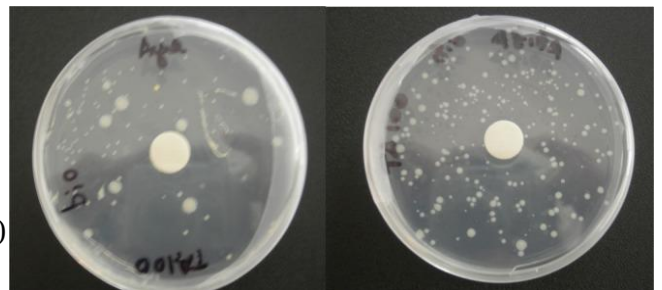
***Indicate the type of mutation present in each, and translate the mutated portion of the sequence into an amino acid sequence in each case. Use the table to determine the amino acid of each codon (5pts)**

		Second letter				
		U	C	A	G	
First letter	U	UUU } Phe UUC } UUA } Leu UUG }	UCU } UCC } Ser UCA } UCG }	UAU } Tyr UAC } UAA Stop UAG Stop	UGU } Cys UGC } UGA Stop UGG Trp	U C A G
	C	CUU } CUC } Leu CUA } CUG }	CCU } CCC } Pro CCA } CCG }	CAU } His CAC } CAA Gln CAG }	CGU } CGC } Arg CGA } CGG }	U C A G
	A	AUU } AUC } Ile AUA } AUG Met	ACU } ACC } Thr ACA } ACG }	AAU } Asn AAC } AAA Lys AAG }	AGU } Ser AGC } AGA } Arg AGG }	U C A G
	G	GUU } GUC } Val GUA } GUG }	GCU } GCC } Ala GCA } GCG }	GAU } Asp GAC } GAA Glu GAG }	GGU } GGC } Gly GGA } GGG }	U C A G

II-3) Study the results of the Ames test below and answer the following question

1) The bacteria used for this test is known as His^- . What does this mean? (1pt)

2) What can you conclude about the chemical tested?(1pt)



Control (no chemical)

Test (with chemical)

3) Why do you see some bacterial colonies grown in the control although no histidine was added? (1pt)